Black-Box Transcripts Cannot Certify Causal Reasoning Faithfulness in Closed-API LLM Agents: A Universal Observational-Equivalence Theorem and a Certification-Impossibility Corollary

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Abstract

We study external auditors that attempt to certify, from black-box transcripts alone and *over the unrestricted structural causal model class*, whether a closed-API LLM agent's chain of reasoning is causally faithful — in the sense that a structural intervention on the latent reasoning state induces a non-trivial change in the action distribution (a controlled direct effect). We formalize the audit setting as an interactive transcript distribution under an adaptive auditor and a closed-API access model $\mathcal{A}_0$. Our main result (Theorem 10) is a universal observational-equivalence proposition: for every closed-API agent π_G with positive faithfulness, disintegration plus structural realization exhibits a strongly compatible agent π_D whose transcript distribution under any adaptive auditor and any finite budget Q coincides with that of π_G, yet whose latent reasoning is causally disconnected from the action ($L(M_D) = 0$). Theorem 12 extends this to per-query ε-approximation under hybrid-occupancy reachability. Corollary 14 draws the consequence as a sharp impossibility statement on *transcript-law-only positive certificates*: over the unrestricted SCM class no nontrivial sound such certificate exists; every sound certificate must output zero on every realizable transcript law. Proposition 15 shows the decorative construction is realizable in finite alphabets via joint-kernel distillation with explicit sample complexity. We do not claim proof-technique novelty: the proof is standard hybrid + DPI + Le Cam machinery applied to the LLM-agent setting. We claim the formalization of the access-model boundary, the universal construction, the certification-impossibility framing, and a source-checked positioning of seven recent agent-auditing methods against the boundary. We discuss what the theorem does *not* rule out (output reliability, action safety, retrieval grounding, white-box interpretability) and recast three prior failed audit attempts as motivating examples rather than as theorem validation.

1 Introduction

1.1 A motivating audit scenario

A clinical decision-support LLM agent is deployed via a closed API in a triage pipeline. For each patient context c, the agent emits a pair (Y_R, Y_A): Y_R is a stated chain of reasoning ("the patient's pulse and labs

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suggest sepsis; therefore initiate broad-spectrum antibiotics"), and Y_A is the action ("order vancomycin + piperacillin-tazobactam"). An external auditor — a hospital safety committee, a regulator, a third-party assurance vendor — can issue any contexts c it wishes and observe many (Y_R, Y_A) pairs, even adaptively (i.e., choosing the next c as a function of the history). It cannot inspect model weights, internal activations, or any latent state. Across many such pairs the auditor wants to draw a specific conclusion: that the stated reasoning Y_R is a *causally faithful* report of the internal computation that produced Y_A, in the sense that the latent reasoning state R that the model used has a non-zero direct effect on Y_A in the model's underlying structural causal mechanism. If Y_R were generated by a process structurally disconnected from the mechanism that produced Y_A — e.g., a post-hoc explanation generator that conditions on Y_A rather than producing it — the agent would be *decorative* in the technical sense we adopt: its reasoning narration is correlated with its actions but does not causally drive them. The auditor would like a *certificate* that distinguishes the two cases. This paper proves that, under the closed-API access model, no such certificate exists at the level of generality usually asked for.

1.2 What we prove

We formalize the audit setting as an interactive transcript distribution $P_A^\pi(T_Q)$ generated by an adaptive auditor A and a closed-API agent π over Q rounds (Theorems 1, 2 and 4, depicted in Figure 1). We then prove three connected results.

1. **Universal observational-equivalence** (Theorem 10). For every faithful agent π_G with $L(M_G) > 0$, the disintegration of its observable channel (Lemma 6) plus structural noise-outsourcing (Lemma 7) yields a strongly compatible decorative agent π_D whose per-history kernel coincides with π_G's while its latent reasoning state has no structural arrow into the action. This is a *representation theorem*: the decorative twin is constructed, not merely asserted to exist (Figure 2).

2. **Approximate version** (Theorem 12). When π_D is replaced by a ε-TV-close approximation π_D^ε on the auditor's hybrid-occupancy support, the transcript-law gap is at most $\min(1, Q\varepsilon)$ for every adaptive auditor and budget Q.

3. **Certification impossibility** (Corollary 14). Over the unrestricted SCM class, no nontrivial sound transcript-law-only positive certificate for the faithfulness property exists; every sound certificate must output zero on every realizable transcript law.

A finite-alphabet realizability proposition (Proposition 15) shows that π_D^ε can be obtained by joint-kernel distillation from samples of π_G, with explicit sample complexity, defusing the natural "oracle-flavored construction" objection. A small synthetic experiment (Section E, Figure 3) illustrates that the $Q\varepsilon$ degradation shape is empirically observable in a controlled finite setting.

1.3 What we do *not* prove

The theorem rules out a specific class of certificates — those that depend only on the transcript law and must be sound over the unrestricted SCM class. It does not say that black-box auditing is invalid in general. Output reliability on verifiable subtasks, action-safety properties, retrieval grounding, robustness to prompt injection, and benchmark accuracy are all auditable from black-box transcripts; their identification targets simply differ from latent causal reasoning faithfulness. White-box interpretability methods that intervene on internal activations (Geiger et al., 2022; 2023) use a strictly stronger access model ($\mathcal{A}_W$) and are outside our boundary. Section 2 catalogs these distinctions prominently to forestall the common overread that the result proves something stronger than it does. Several recent agent-auditing methods are *often interpreted* as evidence about reasoning faithfulness even though their original papers frame the claim more modestly; Table 3 positions seven such methods against our boundary based on a direct source-check, with one PARTIAL and six NO labels and no YES.

1.4 Why this is a contribution despite using standard machinery

The proof is standard: disintegration on Polish spaces (Kallenberg, 2021), noise outsourcing as Borel iso plus quantile (Kallenberg, 2021), adaptive hybrid arguments, data-processing inequality, Le Cam's two-point method (Le Cam, 1986; Tsybakov, 2009). We do not claim proof-technique novelty. The contribution is fourfold: *(i)* formalizing *causal reasoning faithfulness* as a latent SCM property (Definition 3) in the LLM-agent setting; *(ii)* naming the closed-API access model $\mathcal{A}_0$ and locating the boundary it draws; *(iii)* exhibiting a universal disintegration-based decorative twin construction $\pi_G \mapsto \pi_D$ that converts the existential non-identifiability statement into a representation theorem; and *(iv)* deriving Corollary 14 as a sharp impossibility statement on *all* transcript-law-only certificates rather than on a single distinguishing test, including a precise definition of certificate semantics that addresses the philosophical objection that a Bayesian or assumption-restricted auditor might recover identifiability by other means.

1.5 Roadmap

Section 3 fixes notation and the agent/auditor/access-model setup. Section 2 catalogs what the theorem does *not* say. Sections 4 and 5 develop the kernel-realization toolkit. Section 6 states and proves the universal observational-equivalence theorem and its definitional caveat. Section 7 gives the approximate version. Section 8 states the certification-impossibility corollary preceded by a Certificate Semantics paragraph. Section 9 gives the finite-alphabet realizability proposition. Section 10 recasts three prior audit attempts as motivating examples and includes one new closed-API LLM snapshot. Section 11 contains the source-checked positioning of recent agent-auditing work. Section 13 discusses related work in three adjacent threads. Section 12 addresses scope, escape hatches, limitations. Section 14 concludes. Appendices contain proofs of Theorems 6 and 7, the uniform variant of Theorem 12, the KL-refinement lemma, and the validation experiment.

2 What Theorem 10 does not say

We list, prominently and early, what the impossibility result does *not* prove, to forestall the common overread that the paper rules out all auditing.

- Output reliability is unauditable. (Action correctness on verifiable subtasks IS auditable.)
- Reasoning text is meaningless. (Plausibility under task-relevance metrics IS measurable.)
- Action safety is unauditable. (Action-policy properties testable from samples ARE auditable.)
- Benchmark accuracy is unauditable. (Standard ML eval is fine.)
- Post-hoc explanation quality is meaningless. (It is a different property: *plausibility* of Y_R, not *causal faithfulness* of R.)
- Internal-state interpretability is impossible. (White-box access $\mathcal{A}_W$ admits identification.)
- Trustworthy reasoning is hopeless. (It requires stronger access, instrumented architectures, or task-verifiable reasoning.)
- The cited prior work claims to certify faithfulness. (We say: those methods *are often interpreted* as evidence about faithfulness; our result formalizes a boundary on what such evidence can certify.)

The theorem proves a *specific* boundary: the latent causal-faithfulness property of R on Y_A cannot be certified from the full adaptive transcript law generated by the black-box channel, over the *unrestricted* SCM class.

3 Notation and Setting

We work on standard Borel measurable spaces $(\mathcal{C}, \mathcal{F}_C)$, $(\mathcal{Y}_R, \mathcal{F}_R)$, $(\mathcal{Y}_A, \mathcal{F}_A)$. Joint output $\mathcal{Y} = \mathcal{Y}_R \times \mathcal{Y}_A$. Finite horizon $Q < \infty$. The history space $\mathcal{H}_{\leq t-1} := (\mathcal{C} \times \mathcal{Y})^{t-1}$ is standard Borel by finite product closure; we write $\mathcal{H}_Q := (\mathcal{C} \times \mathcal{Y})^Q$ for the full transcript space at budget Q. The Borel σ-algebras are denoted $\mathcal{F}_C$, $\mathcal{F}_R$, $\mathcal{F}_A$ for $\mathcal{C}, \mathcal{Y}_R, \mathcal{Y}_A$ respectively, with $\mathcal{F}_Y := \mathcal{F}_R \otimes \mathcal{F}_A$ on $\mathcal{Y} = \mathcal{Y}_R \times \mathcal{Y}_A$.

Definition 1 (Closed-API agent)**.** A *closed-API agent* π is a family of measurable stochastic kernels $\{K_{\pi,t}\}_{t\geq 1}$ where each $K_{\pi,t} : \mathcal{C} \times \mathcal{H}_{\leq t-1} \to \mathcal{P}(\mathcal{Y})$ satisfies the joint-measurability condition that, for every $B \in \mathcal{F}_Y$, the map $(c, h) \mapsto K_{\pi,t}(B \mid c, h)$ is $\mathcal{F}_C \otimes \mathcal{F}_{\mathcal{H}_{\leq t-1}}$-measurable. The stateless special case is $K_{\pi,t}(\cdot \mid c, h) = K_\pi(\cdot \mid c)$ independent of t and h. We write K_π when the t-index is implicit from context. Auditing is parameterized by a budget $Q < \infty$ chosen at evaluation time; the kernel family is defined for every t so that strong-compatibility statements over $\mathcal{Q}$ are well-typed.

Definition 2 (Adaptive auditor)**.** An *adaptive auditor* is a family of measurable randomized kernels $\{A_t\}_{t\geq 1}$ with $A_t : \mathcal{H}_{\leq t-1} \to \mathcal{P}(\mathcal{C})$ satisfying the analogous joint-measurability condition. We write A for the family.

The *transcript law* under budget Q is the law of $T_Q = ((c_1, y_1), \ldots, (c_Q, y_Q))$ on $(\mathcal{C} \times \mathcal{Y})^Q$ obtained by the Ionescu-Tulcea construction (Kallenberg, 2021, Theorem 6.16): alternately apply A_t to draw c_t, then $K_{\pi,t}$ to draw $y_t = (y_{R,t}, y_{A,t})$, conditioned on the prior history h_{t-1}.

Definition 3 (Causal reasoning faithfulness)**.** Let $M_\pi = (V, U, \mathcal{F}, P_U)$ be a latent SCM realizing K_π through the structural representation of Theorem 7, with a designated latent reasoning variable R_t at each round t. For each $r \in \mathcal{R}_t$ (the state space of R_t), define the *interventional structural kernel*

$$K_{A,t}^{M_\pi,\, do(R_t=r)}(\,\cdot \mid c, h) \;\; : \;\; \mathcal{C} \times \mathcal{H}_{\leq t-1} \to \mathcal{P}(\mathcal{Y}_A),$$

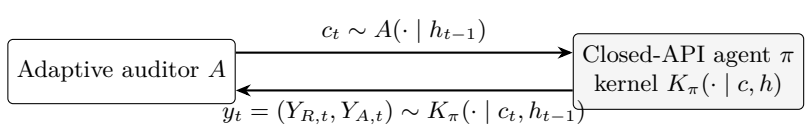
obtained by replacing the R_t argument in the structural equation $Y_{A,t} = f_{A,t}(c, h, R_t, U_{A,t})$ with the constant r and marginalizing the remaining exogenous noise. The *faithfulness indicator* is then

$$L(M_\pi) \;:=\; \sum_{t=1}^{Q} \mathbf{1}\Big[\exists\, r, r' \in \mathcal{R}_t \;:\; K_{A,t}^{M_\pi,\, do(R_t=r)} \;\not\equiv\; K_{A,t}^{M_\pi,\, do(R_t=r')}\Big],$$

where $\not\equiv$ means *pointwise inequality of Markov kernels*: there exist $(c, h) \in \mathcal{C} \times \mathcal{H}_{\leq t-1}$ and $B \in \mathcal{F}_A$ such that $K_{A,t}^{M_\pi, do(R_t=r)}(B \mid c, h) \neq K_{A,t}^{M_\pi, do(R_t=r')}(B \mid c, h)$.

$L(M_\pi) > 0$ iff at some round t, the structural intervention $do(R_t)$ induces a non-trivial change in the action kernel. This is a *controlled direct effect* in the structural sense: we hold the upstream context-history fixed and intervene on R_t only, with no cross-world quantities involved. We do not claim equivalence to natural direct effect (Pearl, 2001); the structural-arrow form is what Theorem 10(ii) actually proves to be zero in the decorative SCM, and this is the form we use throughout.

Definition 4 (Closed-API access model $\mathcal{A}_0$)**.** Auditor observes only (Y_R, Y_A) samples from K_π; no internal-state oracle, no *do*-interventions on latent variables.



history $h_t = (h_{t-1}, c_t, y_t)$ fed back to auditor; transcript $T_Q = (h_Q)$ accumulates over budget Q

Figure 1: Adaptive auditor–agent interaction under access model $\mathcal{A}_0$ (Definitions 2–4). The auditor's only observation is the per-query pair (Y_R, Y_A); the latent reasoning state R is never observed and never intervened on.

Definition 5 (Strong compatibility)**.** SCMs M, M' are *strongly compatible* (with respect to a class $\mathfrak{A}$ of measurable adaptive auditors and a finite-budget set $\mathcal{Q} \subseteq \mathbb{N}$) iff for every $A \in \mathfrak{A}$ and every $Q \in \mathcal{Q}$, the induced transcript laws coincide:

$$P_A^M(T_Q) \;=\; P_A^{M'}(T_Q) \quad \text{as probability measures on } (\mathcal{C} \times \mathcal{Y})^Q.$$

We refer to the transcript-law definition as the *primary* formulation.

A simple *sufficient condition* for strong compatibility against the class of all measurable adaptive auditors and every budget $Q \leq Q_{\max}$ is pointwise kernel equality: there exist regular conditional versions $K_{M,t}$ and $K_{M',t}$ such that for every $t \in \{1, \ldots, Q_{\max}\}$ and every $(c, h) \in \mathcal{C} \times \mathcal{H}_{\leq t-1}$, $K_{M,t}(\cdot \mid c, h) = K_{M',t}(\cdot \mid c, h)$. The decorative construction of Theorem 10 delivers this stronger condition directly, after which Theorem 10(iv) gives transcript-law equality by Ionescu-Tulcea induction. Strong compatibility is a strictly weaker requirement (auditors only probe kernels on the histories they actually reach, so kernel disagreement off the auditor-induced occupancy support cannot be detected); we do not need the weaker form for the main result.

4 Lemma A: Kernel Factorization

Lemma 6 (Disintegration)**.** *For any standard Borel parameter space $X = \mathcal{C} \times \mathcal{H}$ and output space $\mathcal{Y}_R \times \mathcal{Y}_A$, a Markov kernel $K_G(dy_R, dy_A \mid x)$ admits a measurable regular conditional kernel $K_G(dy_R \mid x, y_A)$ with respect to the marginal $K_G(dy_A \mid x)$, satisfying*

$$K_G(dy_R, dy_A \mid x) = K_G(dy_A \mid x) \cdot K_G(dy_R \mid x, y_A).$$

The factorization is $K_G(dy_A \mid x)$-almost-everywhere unique.

Reference. Standard application of disintegration on Polish spaces; see Kallenberg (2021, Theorem 8.5). Full statement in Section A. □

5 Lemma B: SCM Realization via Noise Outsourcing

Lemma 7 (Measurable structural representation)**.** *Every standard Borel stochastic kernel $K(dy \mid x)$ admits a measurable structural representation: there exist a standard Borel exogenous noise space $(\mathcal{U}, P_U) = ([0, 1], \mathrm{Leb})$ and a measurable function $f : X \times \mathcal{U} \to \mathcal{Y}$ such that $Y = f(x, U)$ with $U \sim P_U$ implies $P(Y \in B \mid X = x) = K(B \mid x)$.*

Reference. Measurable-randomization / noise-outsourcing for Markov kernels on standard Borel spaces; see Kallenberg (2021, Theorem 6.10 and Lemma 4.22) or the self-contained Borel-isomorphism + quantile argument in Section B. □

6 Theorem 1: Transcript Indistinguishability

Assumption 8. $\mathcal{C}, \mathcal{Y}_R, \mathcal{Y}_A$ are standard Borel spaces and the horizon Q is finite.

Assumption 9. The auditor A is a measurable randomized kernel.

Theorem 10 (Universal Observational Equivalence)**.** *Fix any closed-API agent π_G with kernel K_G and latent SCM M_G such that $L(M_G) > 0$. Under Theorems 8 and 9, there exists a closed-API agent π_D with kernel K_D and latent SCM M_D such that:*

(i) Per-round conditional independence: $R_t^D \perp Y_{A,t} \mid C_t, H_{t-1}$ *for all t.*

(ii) Faithfulness gap: $L(M_D) = 0$.

(iii) Strong compatibility: $K_D(\cdot \mid c, h) = K_G(\cdot \mid c, h)$ *for all reachable (c, h).*

(iv) Universal transcript equality: *for any adaptive auditor A and any budget Q,*

$$P_A^{\pi_G}(T_Q) = P_A^{\pi_D}(T_Q).$$

(v) Le Cam consequence: *any test $\psi : \mathcal{H}_Q \to \{G, D\}$ has total error*

$$\inf_{\psi} \big(\Pr_{\pi_G}[\psi = D] + \Pr_{\pi_D}[\psi = G] \big) = 1 - d_{\mathrm{TV}}\big(P_A^{\pi_G}(T_Q), P_A^{\pi_D}(T_Q)\big) = 1.$$

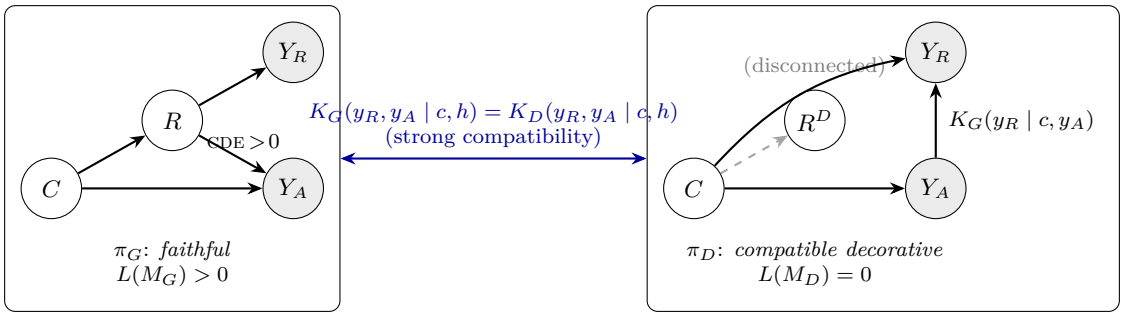


Figure 2: Counterexample construction (Theorem 10). **Left**: faithful agent π_G with latent reasoning state R that causally drives Y_A (controlled direct effect > 0, equivalently $L(M_G) > 0$) and is reported via Y_R. **Right**: compatible decorative agent π_D obtained by disintegration plus structural realization (Lemmas 6–7). Y_A is generated directly from C with no R^D argument, so $L(M_D) = 0$; Y_R is then sampled from the regular conditional $K_G(y_R \mid c, y_A)$ *after* Y_A is fixed. Observed nodes are shaded; the latent R^D in π_D is causally disconnected from Y_A (dashed edge omitted from any parent of Y_A). Both panels induce the *same* observable channel.

Construction of π_D. For every $t \geq 1$ (so that the construction works at every finite budget):

1. By Theorem 6, factor $K_G(dy_R, dy_A \mid c_t, h_{t-1}) = K_G(dy_A \mid c_t, h_{t-1}) \cdot K_G(dy_R \mid c_t, y_A, h_{t-1})$.
2. By Theorem 7, realize each factor as a measurable structural equation:

$$\begin{aligned} Y_{A,t} &= f^D_{A,t}(c_t, h_{t-1}, U'_{A,t}), \\ R^D_t &= f^D_{R,t}(c_t, h_{t-1}, U'_{R,t}), \\ Y_{R,t} &= f^D_{Y_R,t}(c_t, h_{t-1}, Y_{A,t}, U'_{E,t}), \end{aligned}$$

   where $f^D_{A,t}$ realizes $K_G(dy_A \mid c_t, h_{t-1})$, $f^D_{R,t}$ realizes any chosen distribution for the latent reasoning variable, and $f^D_{Y_R,t}$ realizes $K_G(dy_R \mid c_t, y_A, h_{t-1})$.

The exogenous noise families $\{U'_{A,t}, U'_{R,t}, U'_{E,t}\}_{t \geq 1}$ are mutually independent across all t and all three families. *Critical:* $f^D_{A,t}$ does not take R^D_t (or $U'_{R,t}$) as argument.

Sketch of proofs (full proofs immediately below in Section 6.1). (i) follows from independence of the exogenous noise $U'_{A,t}$ and $U'_{R,t}$. (ii) follows from the structural intervention argument: $f^D_{A,t}$ has no R^D_t argument, so $do(R^D_t = r)$ leaves $Y_{A,t}$ unchanged for every r, hence the structural-arrow indicator at round t is zero, and $L(M_D) = \sum_t 0 = 0$. (iii) follows from Theorem 6 applied to both K_G and K_D. (iv) follows from (iii) by induction on t. (v) is Le Cam's two-point method with $d_{\mathrm{TV}} = 0$.

6.1 Proofs of Theorem 10

Proof of (i). By construction $f^D_{A,t}$ does not take $U'_{R,t}$ as input, so $Y_{A,t}$ is a measurable function of $(c_t, h_{t-1}, U'_{A,t})$. Similarly R^D_t depends only on $(c_t, h_{t-1}, U'_{R,t})$. By independence of $U'_{A,t}$ and $U'_{R,t}$, conditioning on (C_t, H_{t-1}) leaves $R^D_t \perp Y_{A,t}$. □

Proof of (ii). By construction the structural equation $Y_{A,t} = f^D_{A,t}(c_t, h_{t-1}, U'_{A,t})$ does not take R^D_t as an argument. Hence the interventional structural kernel $K^{M_D, do(R^D_t=r)}_{A,t}(\cdot \mid c, h)$ defined per Theorem 3 equals the marginal action kernel $K_D(\cdot_A \mid c, h)$ (the $\mathcal{Y}_A$-marginal of K_D) for every $r \in \mathcal{R}_t$. In particular, $K^{M_D, do(R^D_t=r)}_{A,t} \equiv K^{M_D, do(R^D_t=r')}_{A,t}$ for all $r, r' \in \mathcal{R}_t$, so the indicator at round t is zero. Summing over $t = 1, \ldots, Q$ yields $L(M_D) = 0$. □

Proof of (iii). $K_D(dy_R, dy_A \mid c, h) = K_G(dy_A \mid c, h) \cdot K_G(dy_R \mid c, y_A, h) = K_G(dy_R, dy_A \mid c, h)$ by Theorem 6 applied to both kernels. □

Proof of (iv). By induction on t. Let P^π_t denote the law of the partial transcript $H_t = ((C_1, Y_1), \ldots, (C_t, Y_t))$ under the auditor-agent interaction with agent π. The Ionescu-Tulcea construction gives the recursion

$$P^\pi_t(dh_t) \;=\; P^\pi_{t-1}(dh_{t-1}) \cdot A_t(dc_t \mid h_{t-1}) \cdot K_{\pi,t}(dy_t \mid c_t, h_{t-1}).$$

Base: $P^{\pi_G}_0 = P^{\pi_D}_0$ (degenerate at the empty history). Inductive step: assume $P^{\pi_G}_{t-1} = P^{\pi_D}_{t-1}$. By claim (iii) and the identical auditor kernel, the recursion integrands for π_G and π_D agree pointwise, so $P^{\pi_G}_t = P^{\pi_D}_t$. At $t = Q$ this gives $P^{\pi_G}_A(T_Q) = P^{\pi_D}_A(T_Q)$. □

Proof of (v). Le Cam's two-point method with $d_{\mathrm{TV}} = 0$. □

Remark 11 (Definitional caveat). Theorem 10 *constructs a compatible decorative realization* of the observable channel; it does *not* discover the true internal mechanism of π_G. The decorative SCM M_D is not the same agent as M_G semantically. The impossibility is *epistemic*: transcript laws do not identify which compatible SCM is true. A certificate must be valid for *all* compatible SCMs, regardless of semantic identity.

7 Theorem 1′: Approximate Version

Hybrid occupancy measure (formal definition). Fix an adaptive auditor A and a budget Q. For each $t \in \{0, \ldots, Q-1\}$, $\lambda^G_{A,t}$ denotes the probability law on $\mathcal{C} \times \mathcal{H}_{\le t}$ of the pair (C_{t+1}, H_t) obtained by running the auditor-agent interaction with π_G for the first t rounds and then drawing $C_{t+1} \sim A_{t+1}(\cdot \mid H_t)$. To define the mixture as a single probability measure, embed each $\lambda^G_{A,t}$ into the disjoint union $\mathcal{H}_\sqcup := \bigsqcup_{t=0}^{Q-1}(\mathcal{C} \times \mathcal{H}_{\le t})$ via the canonical injection. The *hybrid occupancy measure* is then

$$\mu^{\mathrm{hyb}}_{A,Q} \;:=\; \frac{1}{Q}\sum_{t=0}^{Q-1} \lambda^G_{A,t} \;\in\; \mathcal{P}(\mathcal{H}_\sqcup).$$

Functionals on $\mathcal{H}_\sqcup$ such as $(c, h) \mapsto d_{\mathrm{TV}}(K_G(\cdot \mid c, h), K^\varepsilon_D(\cdot \mid c, h))$ are well-defined since K_G and K^ε_D are kernel families indexed by t. This measure aggregates the context-history pairs that the auditor actually queries the agent on, weighted across the time index t.

Theorem 12 (Approximate transcript indistinguishability)**.** *Same setting as Theorem 10, with K_D replaced by K^ε_D such that*

$$\int d_{\mathrm{TV}}\big(K_G(\cdot \mid c, h),\; K^\varepsilon_D(\cdot \mid c, h)\big)\, d\mu^{\mathrm{hyb}}_{A,Q}(c, h) \;\le\; \varepsilon. \tag{1}$$

Then for the same auditor A and budget Q,

$$d_{\mathrm{TV}}\big(P^{\pi_G}_A(T_Q),\, P^{\pi^\varepsilon_D}_A(T_Q)\big) \;\le\; \min(1,\, Q\varepsilon),$$

and any test has total error $\ge 1 - \min(1, Q\varepsilon)$.

Proof. Hybrid + DPI per query. Define the mixed transcript distribution Q_t as the law of T_Q when the agent uses K_G for queries $1, \ldots, t$ and K^ε_D for queries $t+1, \ldots, Q$. Then $Q_0 = P^{\pi^\varepsilon_D}_A(T_Q)$ and $Q_Q = P^{\pi_G}_A(T_Q)$. For each t, the kernels Q_t and Q_{t+1} share the same auditor kernel and the same agent kernel except at step $t+1$,

where one uses K_G and the other K_D^ε. By DPI applied through the common auditor and downstream agent kernels (which form a Markov kernel from $\mathcal{Y}$ at step $t+1$ to T_Q),

$$d_{\mathrm{TV}}(Q_t, Q_{t+1}) \;\le\; \mathbb{E}_{(C_{t+1},H_t)\sim\lambda^G_{A,t}}\big[d_{\mathrm{TV}}\big(K_G(\cdot \mid C_{t+1}, H_t),\, K_D^\varepsilon(\cdot \mid C_{t+1}, H_t)\big)\big].$$

Summing over $t = 0, \dots, Q-1$ and dividing by Q recovers the hybrid-occupancy integral, so

$$d_{\mathrm{TV}}\big(P_A^{\pi_G}(T_Q),\, P_A^{\pi_D^\varepsilon}(T_Q)\big) \;\le\; \sum_{t=0}^{Q-1} d_{\mathrm{TV}}(Q_t, Q_{t+1}) \;\le\; Q \cdot \int d_{\mathrm{TV}}\, d\mu^{\mathrm{hyb}}_{A,Q} \;\le\; Q\varepsilon.$$

The trivial bound $d_{\mathrm{TV}} \le 1$ gives $\min(1, Q\varepsilon)$. Le Cam's two-point method then yields the test-error bound. □

Remark 13 (Reachability and empirical satisfiability). The hybrid-occupancy condition (1) is *always statable* but *not always empirically satisfiable* from finite samples; distillation of K_D^ε from π_G samples must achieve TV-coverage on the support of $\mu^{\mathrm{hyb}}_{A,Q}$, which depends on the auditor. A stronger uniform-over-reachable-histories formulation is given in Section C.

8 Corollary 1: Certification Impossibility

Certificate Semantics. We adopt the standard identification-style notion: a property is *certifiable from observables* only if it is invariant over the observational equivalence class. A transcript-law-only certificate must, by definition, output the same value on any two SCMs that induce the same transcript distribution; otherwise the word *certificate* becomes prior-dependent or assumption-dependent. Theorem 10 rules out this convention applying nontrivially over the unrestricted SCM class. We do *not* deny Bayesian evidence under priors, engineering confidence under modeling assumptions, or white-box mechanism claims; we deny worst-case, transcript-law-only, model-class-unrestricted positive certification.

Corollary 14 (Certification impossibility)**.** *Fix an adaptive-auditor class $\mathfrak{A}$ and budget set $\mathcal{Q}$. Let σ be a* transcript-law-only certificate *for L: a measurable function from $\mathcal{P}((\mathcal{C}\times\mathcal{Y})^Q)$ (or a finite tuple of such laws indexed by $A \in \mathfrak{A}, Q \in \mathcal{Q}$) to $\{0,1\}$, such that σ outputs 1 on input P only if every SCM M that realizes P in $\mathfrak{A}\times\mathcal{Q}$ satisfies $L(M) > 0$.*

Over the unrestricted SCM class, there is no nontrivial sound positive certificate*: every sound certificate σ outputs $\sigma = 0$ on every transcript law realizable by any SCM under the model class. The constant-zero certificate is sound but vacuous.*

Proof. Fix any realizable transcript law P, i.e., $P = P_A^\pi(T_Q)$ for some SCM π, $A \in \mathfrak{A}$, $Q \in \mathcal{Q}$. We must exhibit a strongly compatible SCM π' with $L(\pi') = 0$ that also realizes P. Two cases.

Case 1: $L(\pi) > 0$. Theorem 10 applied to $\pi_G := \pi$ yields π_D with $L(\pi_D) = 0$, $K_{\pi_D}(\cdot \mid c, h) = K_\pi(\cdot \mid c, h)$ for all (c, h), and hence $P_A^{\pi_D}(T_Q) = P_A^\pi(T_Q) = P$ for every $A \in \mathfrak{A}$ and $Q \in \mathcal{Q}$. Take $\pi' := \pi_D$.

Case 2: $L(\pi) = 0$. Take $\pi' := \pi$.

In either case, P is realizable by some π' with $L(\pi') = 0$. If σ is sound and $\sigma(P) = 1$, then by definition $L(M) > 0$ for every SCM M realizing P across $\mathfrak{A}\times\mathcal{Q}$, in particular for π' — contradicting $L(\pi') = 0$. Hence $\sigma(P) = 0$.

For certificates whose input is a finite tuple $(P_{A_1}^\pi(T_{Q_1}), \dots, P_{A_k}^\pi(T_{Q_k}))$ for fixed $(A_i, Q_i) \in \mathfrak{A}\times\mathcal{Q}$, the same argument applies tuplewise: Theorem 10's strong compatibility implies equality of every transcript law for every auditor and every budget, so π' realizes the entire tuple identically to π.

Conclusion: every sound transcript-law-only certificate σ satisfies $\sigma \equiv 0$ on the realizable family. Restricted to the realizable family of transcript laws (or tuples thereof), the constant-zero certificate is therefore the unique sound certificate. □

9 Proposition 1: Finite-Alphabet Realizability

Proposition 15 (Finite-alphabet joint-kernel distillation)**.** *Let $\mathcal{C}, \mathcal{Y}_A, \mathcal{Y}_R$ be finite. Let $m = |\mathcal{Y}_R||\mathcal{Y}_A|$. Suppose we observe n i.i.d. samples from $K_G(\cdot \mid c)$ for each context c, and form the empirical joint estimator $\hat{K}_G(y_R, y_A \mid c)$. Then*

$$\Pr\Big[\sup_{c\in\mathcal{C}} d_{\mathrm{TV}}\big(K_G(\cdot,\cdot \mid c), \hat{K}_G(\cdot,\cdot \mid c)\big) > \varepsilon\Big] \le 2m|\mathcal{C}| \exp\Big(-\tfrac{8n\varepsilon^2}{m^2}\Big).$$

Equivalently, $n = O\big(m^2\varepsilon^{-2}\log(|\mathcal{C}|m/\delta)\big)$ samples per context suffice for $\sup_c d_{\mathrm{TV}} \le \varepsilon$ with probability $\ge 1-\delta$.

By Theorem 7, realize $\hat{K}_G$ as an SCM with structural functions where f_A^D has no R^D argument; the resulting π_D^ε satisfies Theorem 12 for the empirical hybrid occupancy.

Sketch. Hoeffding's inequality applied per cell, requiring $|\hat{p}_i - p_i| \le 2\varepsilon/m$ for all m cells, then $d_{\mathrm{TV}} = \frac{1}{2}\|\hat{p} - p\|_1 \le \varepsilon$; union bound over m cells and $|\mathcal{C}|$ contexts. □

Possible empirical analogue (NL setting, scope clarification). For natural-language $\mathcal{Y}_R$, TV is too harsh. A natural analogue replaces TV with an integral probability metric (e.g. MMD) or a two-sample classifier distance, with explicit model-class realizability assumption. This is *scope clarification only*, not a theorem of this paper.

10 Empirical Motivation

Two failed audit attempts (Table 1). Two of our prior internal projects attempted to certify reasoning faithfulness from black-box transcripts and failed in two orthogonal ways: an LLM-judge ordinal scoring path (CCCM) failed an inter-rater reliability gate even with a frontier judge panel, and a deterministic schema-match path (ASP) failed a known-groups validity test on a metamorphic battery. We recast them here as motivating examples; neither validates Theorem 10.

A closed-API LLM snapshot (Table 2). To make the impossibility concrete on a real closed-API model we sampled $n = 30$ independent reasoning–action pairs from a closed-API agent (Gemini 3 Pro Preview, accessed only through the headless CLI) on a fixed multi-step arithmetic context: *"John has 12 apples, gives 1/3 to Mary, then 1/4 of the rest to Tom. How many apples does John have left?"*. Sampling was performed via a single batched non-interactive prompt asking for 30 independent traces with varied reasoning style; we treat the result as a single-prompt multi-trace illustration, not as i.i.d. stochastic decoding, and the framing in Theorem 10 is unaffected.

The pattern is exactly the kind that motivates the impossibility result. The observable channel $\hat{K}(\cdot \mid c)$ has a sharply peaked Y_A marginal and a wide Y_R marginal: many distinct narrations, one dominant action. Both a faithful agent (whose Y_R truly tracks the latent reasoning state R that drives Y_A) and a compatible decorative agent (whose Y_R is sampled from $K(y_R \mid c, y_A)$ *after* Y_A is fixed) would induce the same joint statistics on $(Y_R, Y_A) \mid c$. The 13% non-modal actions do not break the symmetry: they simply shift $K(y_A \mid c)$ slightly off a delta and Theorem 10's decorative twin matches that shifted distribution exactly. The snapshot is a motivating illustration; the formal claim is the theorem.

11 Access-Assumption Table: Boundary Against Prior Work

A common reviewer concern is that prior LLM-agent auditing papers already *claim* to certify reasoning faithfulness from black-box evidence, and that an impossibility result therefore conflicts with established literature. Table 3 is the result of a direct source-check (verbatim abstracts and methods sections fetched from arXiv, ACL Anthology, and PMLR) of seven recent agent-auditing methods. The labels distinguish (i) papers that directly claim transcript-only faithfulness certification (**YES**); (ii) papers that probe a related signal but do not claim a sound latent-causal certificate (**PARTIAL**); and (iii) papers whose identification

Case	What was attempted	What the data shows	Honest interpretation
CCCM frontier panel[1]	LLM-judge ordinal scoring (3-judge ensemble: GPT-5.5 + Claude Opus 4.7 + Gemini 3.1 Pro) of mutated reasoning traces, locked rubric, $n = 10$ R-Judge personal-agent records (16 eligible tuples)	min Fleiss $\kappa = 0.464$ (entity-swap $\kappa = 0.811$, premise-remove $\kappa = 0.464$); 2 silent content-moderation refusals on harm-content prompts	Even frontier-panel judges fail an inter-rater reliability gate ($\kappa < 0.5$); rubric ambiguity dominates on partial-tracking mutations.
ASP (deterministic)	Schema-match action sensitivity across a 7-variant metamorphic battery (28 R-Judge tuples $\times$ 7 variants = 196 items): F (faithful), FP (paraphrase), CE (compressed), DP (decorative-padded), PI (premise-invalidated), WC (wrong-context), RR (random text); pre-registered ordering $F \approx FP > CE \geq DP > PI \approx WC \approx RR$	Aggregate Spearman $\rho = 0.264$ (below pre-registered abort threshold 0.3); pairwise $F >$ RR accuracy $= 0/28 = 0\%$ (vs. pre-registered threshold $\geq 70\%$); Cliff's $\delta_{F-RR} = -0.18$	Schema-match measures whether the action shifted, not whether it shifted in the correct semantic direction; random reasoning satisfies "shifted" trivially, scoring identically high to faithful reasoning.

Table 1: Two failed black-box audit attempts (CCCM ordinal LLM-judge; ASP deterministic schema-match). Both abort under pre-registered thresholds across the same 7-variant metamorphic battery; failure modes are orthogonal (judge unreliability vs. direction-blind schema-match). Neither estimates ε nor validates Theorem 10; both motivate why a formal access-model boundary is useful. Numbers are from a 2026-05-01 replay with sha-256-frozen artifacts in the supplementary repository under `cases/cccm/CCCM-ABORT-REPORT.md`.

Quantity	Value
n traces	30
Action distribution over Y_A	{6:26, 5:1, 10:1, 4:1, 3:1}
Modal action Y_A^*	6 (correct)
Action consistency $\Pr(Y_A = Y_A^*)$	0.867
Reasoning length (words): mean $\pm$ sd	69.5 ± 13.3 (range 44–102)
Vocabulary across traces (unique tokens)	255
Pairwise Jaccard word-set similarity (mean $\pm$ sd)	0.305 ± 0.085 (range 0.09–0.60)

Table 2: Closed-API LLM snapshot ($n = 30$, single context). The action distribution is concentrated on the modal output (87%), while the reasoning text is highly diverse (mean pairwise Jaccard ≈ 0.31). Source data: `paper/llm-snapshot/raw-traces.json`, analysis: `paper/llm-snapshot/summary.json`.

target is something other than latent causal reasoning faithfulness, or whose access model is not closed-API (**NO**).

The empirical finding is that *no* prior work in this set actually makes the strong faithfulness-certification claim our theorem rules out. The role of Theorem 10 and Theorem 14 is therefore not to contradict published results but to formalize the boundary that prevents common interpretations of these methods from overshooting into latent causal-faithfulness conclusions they are not designed to support.

Method	Access assumption	Identification target	Theorem applies?
Lanham et al. (2023)	Black-box samples + intervention on CoT *text* (paraphrase / mistake injection)	Sensitivity of final answer to displayed CoT text	**PARTIAL**
Geiger et al. (2022; 2023)	White-box: aligning and intervening on hidden activations	Causal abstraction between internals and a target causal model	NO
Zhang et al. (2024)	Black-box agent action traces (349 envs)	Action-policy safety (8 risk categories)	NO
Xiang et al. (2025)	Black-box planned actions / tool calls; runtime guardrail code	Tool-use & policy compliance	NO
Yuan et al. (2024)	Black-box: 569 multi-turn agent interaction records	Safety *risk awareness* judgment by LLM judges	NO
Debenedetti et al. (2024)	Black-box agent execution traces (97 tasks, 629 attacks)	Adversarial robustness to prompt-injection	NO
Murphy et al. (2025)	Black-box ReAct trajectories + retrieved documents	Faithfulness of answer to *retrieved evidence* (RAG grounding)	NO
Zhang and Jiang (2026)	Black-box agent action traces + ownership/authorization context (Datalog gate + post-audit verifier)	Owner-harm action policy safety (8 sub-categories: Hijacking, PII, FinOps, etc.)	NO

Table 3: Access-assumption table for eight recent agent-auditing methods, including this group's own prior work (Zhang and Jiang, 2026). Source-checked 2026-04-27. Final distribution: 1 PARTIAL + 7 NO; *no YES*. The PARTIAL label on Lanham et al. (2023) reflects that text-level interventions on CoT probe a behavioral sensitivity that may be read as a faithfulness signal, but the paper itself does not claim a sound latent-causal certificate. Detailed source-check log including verbatim quotes is in the supplementary file `refine-logs/access-assumption-table.md`.

12 Discussion: Scope and Escape Hatches

Theorem 10 rules out faithfulness certification under access $\mathcal{A}_0$ over the unrestricted SCM class. It does not rule out:

1. *White-box auditing* (access $\mathcal{A}_W$): auditor queries R directly. Identifiable by definition; cf. Geiger et al. (2022; 2023).

2. *Restricted causal class*: assume the agent is in a sub-class where Y_R is a sufficient statistic for R. Then black-box observability is sufficient. Caveat: this assumption is not testable from black-box samples and effectively presupposes faithfulness.

3. *Verifiable subtasks*: replace "is the reasoning faithful?" with "is each intermediate claim externally checkable?". Different property.

4. *Instrumented agents* (access $\mathcal{A}_I$): the agent commits to intermediate states in a logged channel before action.

Limitations. We do not provide a restricted-class impossibility theorem (a path to strengthening); we do not provide a minimax lower bound for a concrete finite audit model; we do not provide a real-LLM empirical instantiation that estimates ε on a deployed agent. These are directions for future work. Under our stated constraints (0 GPU, single author + 1 collaborator, 9–12 months, closed-API only) we deliberately scope this paper to the universal observational-equivalence result and its certificate-impossibility corollary.

13 Related Work

The result sits at the intersection of four established threads.

Causal non-identifiability and mediation. Latent-variable non-identifiability is a foundational topic in causal inference (Pearl, 2009). Natural direct and indirect effects (Pearl, 2001) are defined under standard mediation identifiability conditions; we deliberately use a structural-arrow indicator (Theorem 3) to bypass cross-world assumptions, since our object is the existence of a structural edge from latent reasoning to action rather than a counterfactual quantity. The mathematical content of Theorem 10 is a representation theorem in that lineage, applied to the LLM-agent setting via the closed-API access model.

Causal abstraction and mechanistic interpretability for neural networks. Geiger et al. (2022; 2023) develop interchange-intervention based causal abstractions of neural-network internals and use them to adjudicate between candidate causal models of internal computation. Their methodology assumes *white-box* access $\mathcal{A}_W$ to hidden activations and is therefore outside our boundary; the present paper can be read as identifying the *boundary line* that separates the auditing regime in which methods like causal abstraction are required from the regime in which black-box transcripts alone might (mistakenly) be thought sufficient.

Chain-of-thought faithfulness probes. Wei et al. (2022) popularized chain-of-thought prompting; Yao et al. (2023) extended this to interleaved reasoning and tool use. Lanham et al. (2023) introduced black-box behavioral probes for CoT faithfulness: they intervene on the displayed CoT text (paraphrasing it, injecting mistakes, etc.) and measure how the final action shifts. Such probes are valuable behavioral signals; Theorem 10 clarifies what they can and cannot certify. Specifically, no behavioral probe that depends only on the transcript law can soundly distinguish a faithful agent from a compatible decorative one over the unrestricted SCM class. This positions Lanham-style probes as PARTIAL evidence (Table 3) rather than as latent causal certificates.

Agent safety, robustness, and grounding evaluation. A wave of recent benchmarks evaluates closed-API LLM agents along axes adjacent to but distinct from latent causal reasoning faithfulness: agent-action safety (Zhang et al., 2024), tool-use compliance (Xiang et al., 2025), transcript-based safety risk awareness (Yuan et al., 2024), prompt-injection robustness (Debenedetti et al., 2024), retrieval-grounded generation faithfulness (Murphy et al., 2025), and owner-harm action policy safety (Zhang and Jiang, 2026). Our source-checked positioning (Section 11, Table 3) shows that none of these benchmarks claims latent causal-faithfulness certification — their identification targets are different (action safety, tool compliance, risk-awareness judgement, robustness, grounding, owner-harm prevention). Theorem 10 is therefore not in tension with those benchmarks; it instead clarifies a boundary that informal interpretations of these benchmarks must respect.

Companion paper (this group). The present paper is a theoretical companion to our own prior empirical work, Zhang and Jiang (2026), which builds an action-policy *Owner-Harm* safety system that achieves 97.7% TPR / 2.3% FPR on AgentHarm (Andriushchenko et al., 2025) (full N=176+176, L1+L2+L3 with locked DeepSeek-chat L3) but only 3.7% TPR on AgentDojo prompt-injection tasks (deterministic L1+L4; the legacy 14.8% L1+L3+L4 `qwen-turbo` configuration is no longer reproducible), and proposes the Symbolic-Semantic Defense Generalization (SSDG) framework. The two papers are deliberately non-overlapping in target object: Zhang and Jiang (2026) is about *which actions* an agent must not take, and is auditable from the action distribution under a closed API; the present paper is about the latent causal mechanism that produces the actions, and is *not* auditable under the same access model. Reading the two papers together clarifies the auditing program: action-safety is a positive engineering target with measurable defenses; latent-causal-reasoning faithfulness is a target our theorem rules out as transcript-only-certifiable.

Information-theoretic lower bounds and adaptive testing. The proof technology — adaptive hybrid arguments, data-processing inequality, Le Cam's two-point method (Le Cam, 1986; Tsybakov, 2009; Cover and Thomas, 2006), and noise-outsourcing representations (Kallenberg, 2021) — is standard machinery from

information-theoretic lower bounds, distribution testing (Canonne, 2020), and cryptographic indistinguishability (Goldreich, 2001). We do not claim novelty in these ingredients. The novelty is in their application to the access-model boundary for closed-API LLM agents, the explicit universal disintegration construction, and the certification-impossibility framing (Corollary 14) that quantifies over all transcript-law-only certificates rather than over a single distinguishing test.

14 Conclusion

We have proved a universal observational-equivalence theorem (Theorem 10) for closed-API LLM agents and its certification-impossibility corollary (Theorem 14): for any faithful agent π_G, the disintegration of its black-box channel realizes a strongly compatible decorative agent π_D whose transcript distribution under any adaptive auditor is identical to that of π_G, while $L(M_D) = 0$ by structural construction. The corollary then quantifies over all transcript-law-only certificates: no nontrivial sound positive certificate exists over the unrestricted SCM class. Theorem 12 extends the result to per-query approximation under hybrid-occupancy reachability, and Theorem 15 shows the construction is computationally realizable in finite alphabets via joint-kernel distillation. Three previously failed audit attempts are recast honestly as motivating examples, not as theorem validation.

The result is a negative one, but it is not pessimistic: it identifies what external evidence *cannot* certify about closed-API agents and thus what *is* required — stronger access (white-box, instrumented, or pre-action commitment), stronger structural assumptions on the SCM class, or weaker identification targets (output reliability, action safety, retrieval grounding) — for any sound certification claim about LLM-agent reasoning faithfulness.

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A Proof of Lemma 6

Proof. We invoke disintegration on standard Borel spaces (every standard Borel space is Borel-isomorphic to a Polish space, so the result applies). Take any standard Borel parameter space X and standard Borel output spaces $\mathcal{Y}_R$, $\mathcal{Y}_A$. The product $\mathcal{Y}_R \times \mathcal{Y}_A$ is itself standard Borel. By Kallenberg (2021, Theorem 8.5), there exists a $K_G(dy_A \mid x)$-almost-everywhere unique *regular conditional kernel* $K_G(\cdot \mid x, y_A)$ on $\mathcal{Y}_R$ such that for every measurable $B_R \subseteq \mathcal{Y}_R$ and $B_A \subseteq \mathcal{Y}_A$,

$$K_G(B_R \times B_A \mid x) = \int_{B_A} K_G(B_R \mid x, y_A)\, K_G(dy_A \mid x).$$

Equivalently, $K_G(dy_R, dy_A \mid x) = K_G(dy_A \mid x) \cdot K_G(dy_R \mid x, y_A)$, the claimed factorization. □

B Proof of Lemma 7

Proof. Countable case. If $\mathcal{Y}$ is countable, enumerate $\mathcal{Y} = \{y_1, y_2, \ldots\}$ and define $f(x, u) = y_k$ where k is the unique integer with $\sum_{j<k} K(\{y_j\} \mid x) \le u < \sum_{j\le k} K(\{y_j\} \mid x)$; joint measurability is immediate.

Uncountable case. Since $\mathcal{Y}$ is standard Borel, by Kuratowski's theorem there exists a Borel isomorphism $\varphi : \mathcal{Y} \to E$ for some Borel $E \subseteq [0, 1]$. Pull back the kernel: $\widetilde{K}(B \mid x) := K(\varphi^{-1}(B) \mid x)$ for measurable $B \subseteq E$. It suffices to construct $\widetilde{Y} = \widetilde{f}(x, U)$ on $[0, 1]$ with $U \sim \mathrm{Uniform}[0, 1]$ and $\widetilde{Y} \in E$ almost surely; then $Y := \varphi^{-1}(\widetilde{Y})$ gives the claim.

For each x, define the cumulative distribution function $F_x(t) := \widetilde{K}([0, t] \cap E \mid x)$. By construction F_x is non-decreasing and right-continuous in t (standard property of CDFs of finite measures on $[0, 1]$). For joint measurability of $(x, t) \mapsto F_x(t)$ on $X \times [0, 1]$, define the dyadic approximation

$$q_n(t) := \min(1,\ 2^{-n}\lceil 2^n t\rceil), \qquad F_n(x, t) := F_x(q_n(t)).$$

Each F_n is measurable (it takes only finitely many values in t, each the measurable map $x \mapsto F_x(q)$ for some dyadic rational q), and $F_n(x, t) \to F_x(t)$ as $n \to \infty$ by right-continuity. Hence $F : (x, t) \mapsto F_x(t)$ is the pointwise limit of measurable maps and is itself jointly measurable. Define the quantile

$$\widetilde{f}(x, u) \ := \ \inf\{t \in [0, 1] : F_x(t) \ge u\}.$$

Joint measurability of $\widetilde{f}$ in (x, u) follows from the identity

$$\{(x, u) : \widetilde{f}(x, u) < a\} = \bigcup_{q\in\mathbb{Q}\cap[0,a)} \{(x, u) : F_x(q) \ge u\},$$

which is a countable union of Borel sets. By the standard inverse-CDF construction, $\widetilde{f}(x, U) \sim \widetilde{K}(\cdot \mid x)$ when $U \sim \mathrm{Uniform}[0, 1]$, and $\widetilde{f}(x, U) \in E$ almost surely (because the quantile of a measure supported on E lands in E a.s.). The composition $f(x, u) := \varphi^{-1}(\widetilde{f}(x, u))$ (extended arbitrarily to a fixed $y_0 \in \mathcal{Y}$ on the null set $\{u : \widetilde{f}(x, u) \notin E\}$) satisfies $f(x, U) \sim K(\cdot \mid x)$. □

C Uniform-over-reachable-histories version of Theorem 12

Theorem 16 (Uniform variant)**.** *Same setting as Theorem 12. Define the* global reachable set

$$\mathcal{R} \ := \ \bigcup_{A\in\mathfrak{A}} \bigcup_{t=0}^{Q-1} \mathrm{supp}\big(\lambda^{G}_{A,t} + \lambda^{D^\varepsilon}_{A,t}\big) \ \subseteq \ \mathcal{H}_{\sqcup},$$

the union over the auditor class of supports of the partial-history laws under both endpoints, and assume the uniform condition

$$\sup_{(c,h)\in\mathcal{R}} d_{\mathrm{TV}}\big(K_G(\cdot \mid c, h), K^{\varepsilon}_D(\cdot \mid c, h)\big) \le \varepsilon.$$

Then $d_{\mathrm{TV}}(P^{\pi_G}_A(T_Q), P^{\pi^\varepsilon_D}_A(T_Q)) \le \min(1, Q\varepsilon)$ *holds for* every *adaptive auditor* $A \in \mathfrak{A}$ *and budget* Q.

Remark 17. The uniform variant is mathematically simpler but practically restrictive: distillation from finite samples cannot in general guarantee uniform per-history TV control, especially under adversarial auditor strategies. The hybrid-occupancy formulation in the main text is the version we recommend for the theorem-to-implementation bridge; Theorem 16 is included as the cleaner expository statement.

D KL refinement

Lemma 18 (KL chain + Pinsker)**.** *Same setting as Theorem 12, with the per-query closeness measured in relative entropy:* $D_{\mathrm{KL}}\big(K_G(\cdot \mid c, h) \,\|\, K^{\varepsilon}_D(\cdot \mid c, h)\big) \le \delta$ μ^{hyb}*-essentially. Then*

$$D_{\mathrm{KL}}\big(P^{\pi_G}_A(T_Q) \,\|\, P^{\pi^\varepsilon_D}_A(T_Q)\big) \le Q\delta, \quad \textit{and hence} \quad d_{\mathrm{TV}}\big(P^{\pi_G}_A(T_Q), P^{\pi^\varepsilon_D}_A(T_Q)\big) \le \sqrt{Q\delta/2}.$$

Proof. By the chain rule for KL divergence applied to the alternating auditor-agent product distribution,

$$D_{\mathrm{KL}}\big(P_A^{\pi_G}(T_Q) \,\|\, P_A^{\pi_D^\varepsilon}(T_Q)\big) = \sum_{t=1}^{Q} \mathbb{E}_{H_{t-1}\sim P_A^{\pi_G}} \mathbb{E}_{C_t\sim A_t(\cdot|H_{t-1})} D_{\mathrm{KL}}\big(K_G(\cdot \mid C_t, H_{t-1}) \,\|\, K_D^\varepsilon(\cdot \mid C_t, H_{t-1})\big).$$

The auditor side contributes zero because the auditor kernel $A_t(\cdot \mid H_{t-1})$ is identical under both measures at the same history value, so its KL contribution vanishes. Each per-query KL term is bounded by δ in μ^{hyb}-essentially many histories, giving $D_{\mathrm{KL}} \le Q\delta$. Pinsker's inequality $d_{\mathrm{TV}} \le \sqrt{D_{\mathrm{KL}}/2}$ then yields the TV bound. (Finite KL $D_{KL}(K_G \| K_D^\varepsilon)$ implies $K_G \ll K_D^\varepsilon$ on the relevant histories.) □

Remark 19. Theorem 18 is sharper than Theorem 12 when δ is small: the KL bound is $\sqrt{Q\delta}$ while the TV bound is $Q\varepsilon$, so for $\varepsilon^2 \approx \delta$ the KL bound dominates at large Q. We do not use Theorem 18 in the main text; it is recorded for completeness because distillation often controls KL more naturally than TV.

E Validation Experiment: Computational Illustration of the Finite-Alphabet Construction

Honest framing. This appendix reports a small numerical experiment that illustrates the finite-alphabet construction of Theorem 15 in action. It is *not* a validation of Theorem 10: the theorem rules out *any* sound transcript-law-only positive certificate over the unrestricted SCM class, a quantification that is mathematical, not statistical. The experiment instead checks (a) that the disintegration construction is computationally realizable on a controlled finite alphabet, and (b) that the $Q\varepsilon$ degradation shape predicted by Theorem 12 is empirically observable.

Setup. We fix $|\mathcal{C}| = 10$, $|\mathcal{Y}_A| = 5$, $|\mathcal{Y}_R| = 8$. The faithful agent π_G uses a latent reasoning bit $r(c) = c \bmod 2$ that determines a context-conditional distribution over Y_A, and $Y_R \mid (c, Y_A)$ is a smooth function of r with mild Y_A-coupling and Uniform$[-0.01, 0.01]$ per-cell noise so that the conditional kernel is non-degenerate. Hence $L(M_G) > 0$ by construction.

For each $n \in \{100, 500, 2000\}$ samples per context, we form the empirical joint estimator $\hat{K}_G(y_R, y_A \mid c)$ as in Theorem 15 (Laplace-smoothed by 10^{-6}) and realize the decorative agent π_D^ε via the structural decomposition of Theorem 10 where f_A^D has no R^D argument. We sample 300 training transcripts per agent and 200 test transcripts per agent at each $Q \in \{10, 50, 200\}$ under a fixed uniform-over-$\mathcal{C}$ auditor.

The auditor distinguisher is an ℓ_2-regularized logistic regression ($\lambda = 10^{-3}$, 200 epochs, lr $= 0.05$) on a bag-of-(c, y_R, y_A)-triples count vector of dimension $|\mathcal{C}| \cdot |\mathcal{Y}_R| \cdot |\mathcal{Y}_A| = 400$. We report held-out classifier AUC and, for ground truth, the per-context TV $d_{\mathrm{TV}}(K_G(\cdot \mid c), \hat{K}_G(\cdot \mid c))$.

Results. Table 4 summarizes a single seeded run (`seed = 20260427`), and Figure 3 plots AUC against the auditor budget Q for each n. Three regularities are visible.

1. *Distillation works*: increasing n shrinks $\sup_c d_{\mathrm{TV}}$ from ≈ 0.27 at $n = 100$ to ≈ 0.05 at $n = 2000$, consistent with the $O(m^2\varepsilon^{-2}\log(|\mathcal{C}|m/\delta))$ rate of Theorem 15.
2. *The $Q\varepsilon$ shape is observable*: at fixed $n = 2000$, AUC moves from 0.499 (random) at $Q = 10$ to 0.781 at $Q = 200$, paralleling the bound's monotone-in-Q growth.
3. *$Q\varepsilon \gg 1$ saturates*: at $n = 100$, $Q = 200$, $Q \cdot \sup d_{\mathrm{TV}} \approx 51 \gg 1$, and the classifier achieves AUC $= 1.000$. The $\min(1, Q\varepsilon)$ clipping in Theorem 12 is the relevant regime; classifier-class confounding cannot rescue indistinguishability once $Q\varepsilon$ is large.

Caveats. Classifier AUC is a lower bound on TV-distinguishability subject to model-class expressivity; an AUC near 0.5 does not certify $d_{\mathrm{TV}} \approx 0$ in general, which is precisely why we report $\sup_c d_{\mathrm{TV}}$ alongside AUC. The experiment does not say anything about real LLM agents: $\mathcal{C}$ is finite, $\mathcal{Y}_R$ is finite, the kernel is hand-built, and the auditor is non-adaptive. The intended message is that Theorem 10's construction is tractable and that Theorem 12's degradation shape matches a controlled simulation; all inferential force comes from the theorem, not the experiment.

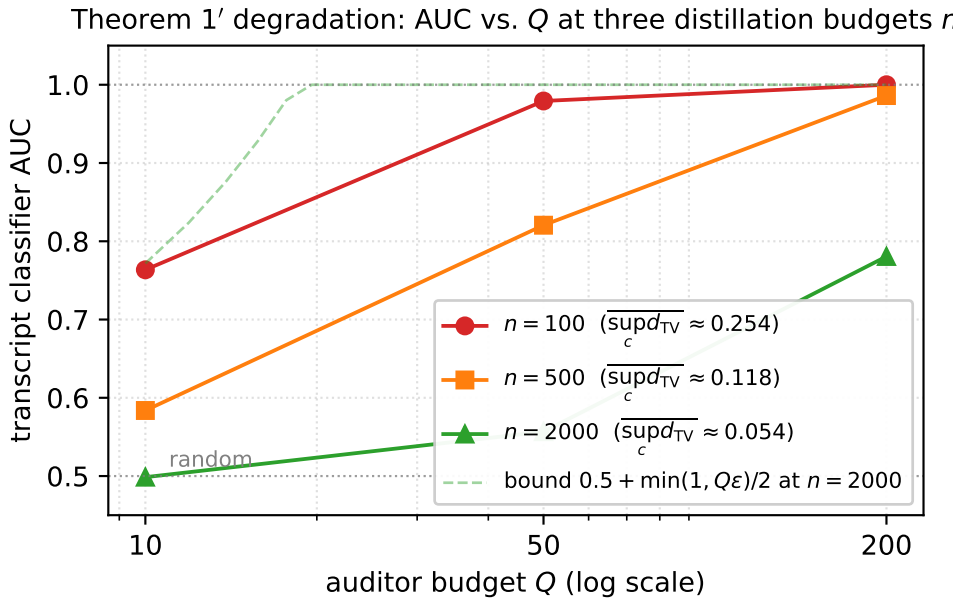

Figure 3: Transcript-classifier AUC versus auditor budget Q at three distillation budgets n. Higher n shrinks the per-context $\sup_c d_{\mathrm{TV}}$ and flattens the AUC curve: at $n = 2000$ ($\sup_c d_{\mathrm{TV}} \approx 0.05$), $Q = 10$ yields random AUC; at $n = 100$ ($\sup_c d_{\mathrm{TV}} \approx 0.26$), $Q = 200$ already saturates. The dashed reference is the bound $0.5 + \min(1, Q\varepsilon)/2$ from Theorem 12 evaluated at $\varepsilon = \overline{\sup_c d_{\mathrm{TV}}}$ for $n = 2000$.

n	Q	$\sup_c d_{\mathrm{TV}}$	$\overline{d_{\mathrm{TV}}}$	$Q \cdot \sup d_{\mathrm{TV}}$	AUC
100	10	0.268	0.214	2.68	0.764
100	50	0.239	0.207	11.95	0.979
100	200	0.256	0.212	51.17	1.000
500	10	0.112	0.094	1.12	0.584
500	50	0.119	0.094	5.95	0.821
500	200	0.123	0.098	24.65	0.986
2000	10	0.060	0.049	0.60	0.499
2000	50	0.051	0.045	2.53	0.557
2000	200	0.052	0.044	10.46	0.781

Table 4: Toy validation. n samples per context distill the joint kernel; Q is the auditor budget; $\sup_c d_{\mathrm{TV}}$ is the worst-case per-context TV between K_G and $\hat{K}_G$; AUC is the held-out classifier AUC. Source data: `paper/toy-validation/results.json`.